

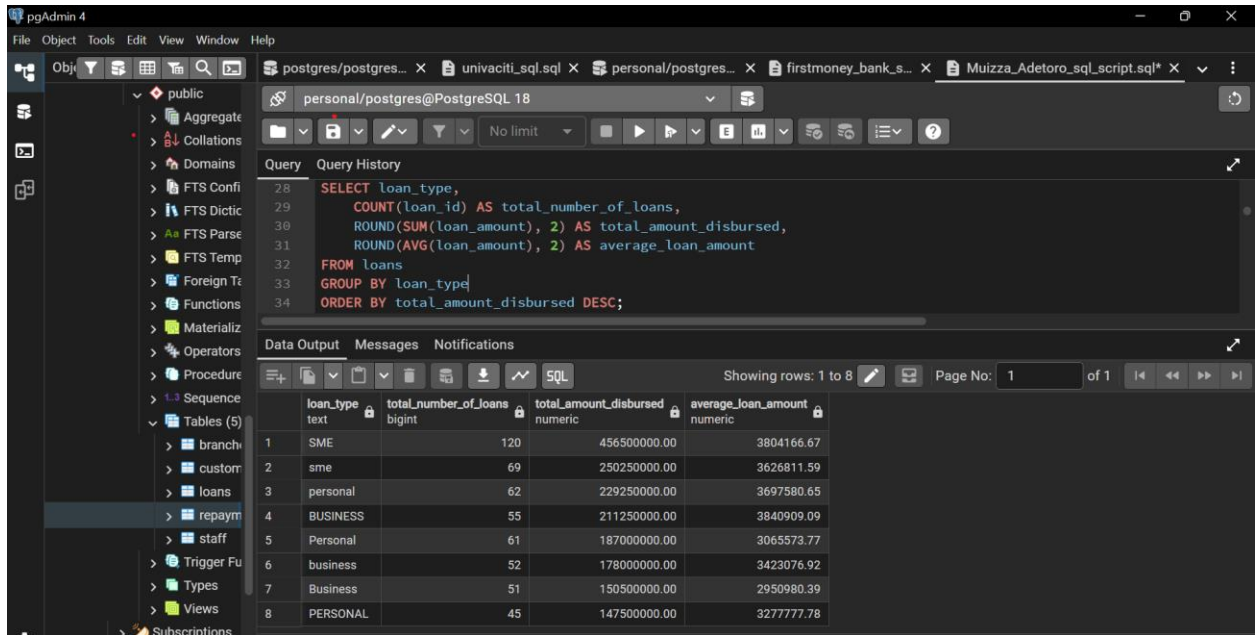
Loan Performance Analysis Query Result

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Q1: Loan Summary by Type

Using the loans table only, return each loan_type with: Total number of loans, Total amount disbursed (rounded to 2 decimal places), Average loan amount. Sort by total amount disbursed from highest to lowest.



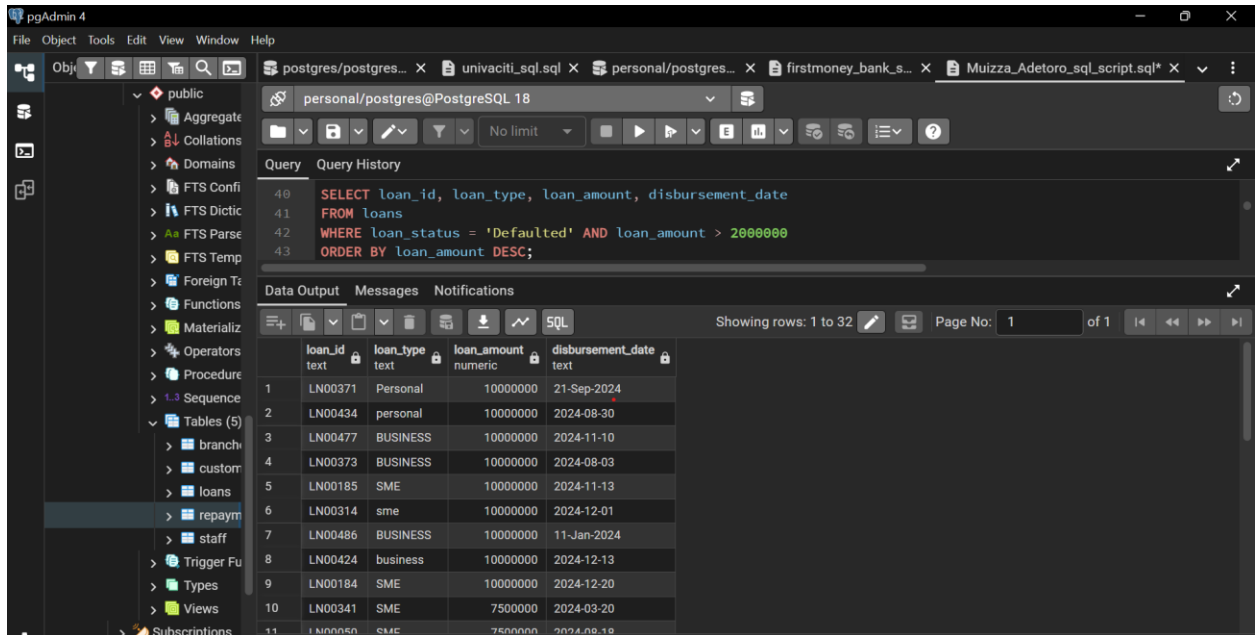
The screenshot shows the pgAdmin 4 interface. The left sidebar displays the database structure, including the 'loans' table under the 'public' schema. The central pane shows a SQL query that aggregates data from the 'loans' table by 'loan_type'. The query calculates the total number of loans, the total amount disbursed (rounded to two decimal places), and the average loan amount for each loan type. The results are displayed in a table with 8 rows, sorted by total amount disbursed in descending order.

```
SELECT loan_type,
COUNT(loan_id) AS total_number_of_loans,
ROUND(SUM(loan_amount), 2) AS total_amount_disbursed,
ROUND(AVG(loan_amount), 2) AS average_loan_amount
FROM loans
GROUP BY loan_type
ORDER BY total_amount_disbursed DESC;
```

loan_type	total_number_of_loans	totalAmountDisbursed	average_loan_amount
SME	120	456500000.00	3804166.67
sme	69	250250000.00	3626811.59
personal	62	229250000.00	3697580.65
BUSINESS	55	211250000.00	3840909.09
Personal	61	187000000.00	3065573.77
business	52	178000000.00	3423076.92
Business	51	150500000.00	2950980.39
PERSONAL	45	147500000.00	3277777.78

Q2: High-Value Defaulted Loans

Return loan_id, loan_type, loan_amount, and disbursement_date for all loans where loan_status = 'Defaulted' AND loan_amount > 2,000,000. Sort by loan_amount descending.



The screenshot shows the pgAdmin 4 interface. The left sidebar displays the database structure, with 'Tables (5)' expanded under the 'public' schema. The 'loans' table is selected. The main pane shows a SQL query in the 'Query' tab:

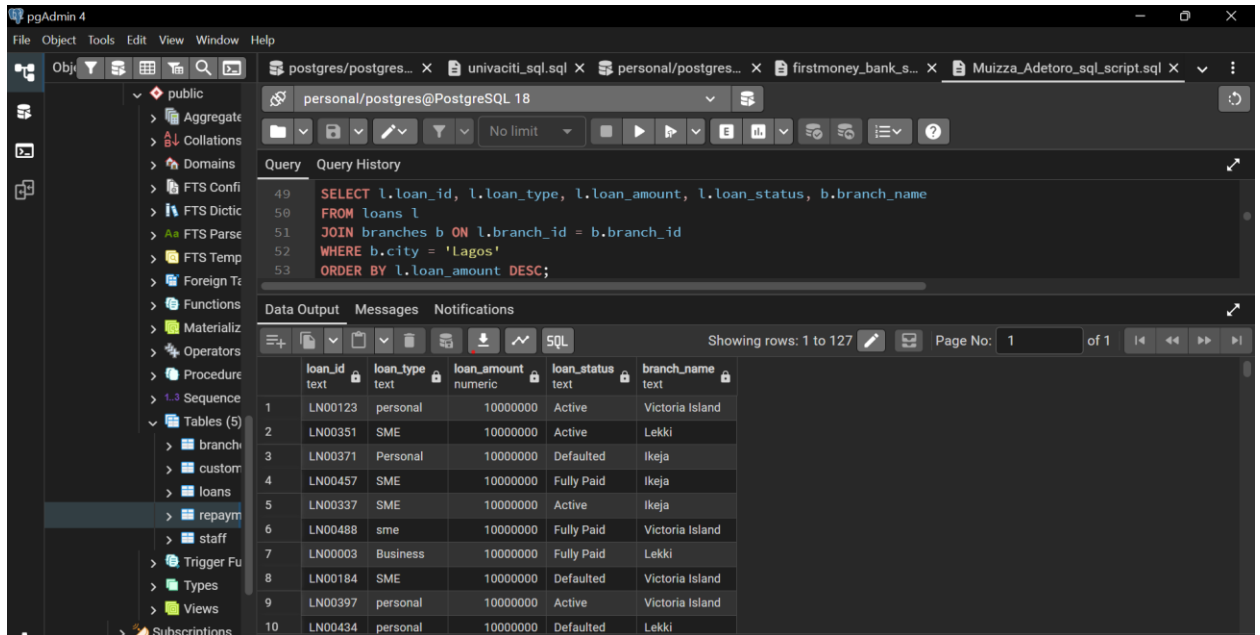
```
40 SELECT loan_id, loan_type, loan_amount, disbursement_date
41 FROM loans
42 WHERE loan_status = 'Defaulted' AND loan_amount > 2000000
43 ORDER BY loan_amount DESC;
```

The 'Data Output' tab shows the results of the query, displaying 10 rows of data. The columns are loan_id, loan_type, loan_amount, and disbursement_date. The results are sorted by loan_amount in descending order.

	loan_id	loan_type	loan_amount	disbursement_date
1	LN00371	Personal	10000000	21-Sep-2024
2	LN00434	personal	10000000	2024-08-30
3	LN00477	BUSINESS	10000000	2024-11-10
4	LN00373	BUSINESS	10000000	2024-08-03
5	LN00185	SME	10000000	2024-11-13
6	LN00314	sme	10000000	2024-12-01
7	LN00486	BUSINESS	10000000	11-Jan-2024
8	LN00424	business	10000000	2024-12-13
9	LN00184	SME	10000000	2024-12-20
10	LN00341	SME	7500000	2024-03-20

Q3: Lagos Region Loans with JOIN

Return loan_id, loan_type, loan_amount, loan_status, and branch_name for every loan disbursed from a branch in Lagos. You must JOIN the loans and branches tables. Sort by loan_amount descending.



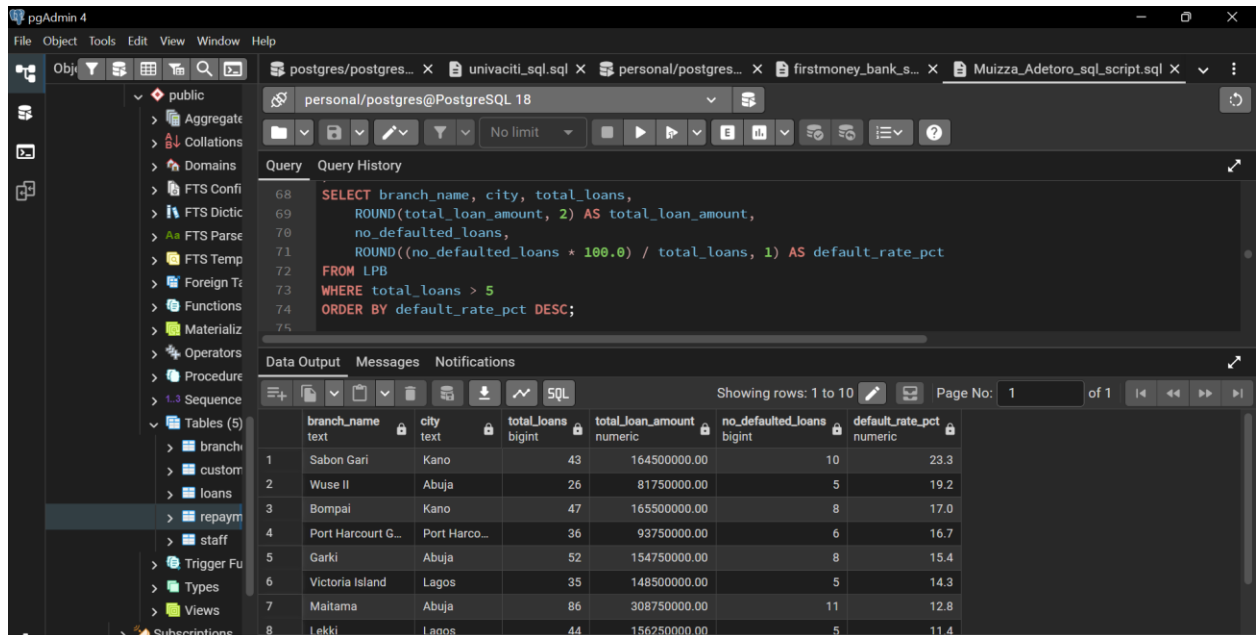
The screenshot shows the pgAdmin 4 interface. The left sidebar displays the database structure, including the 'loans' table. The central pane shows a SQL query that joins the 'loans' table with the 'branches' table, filtering for loans in Lagos and sorting by loan amount in descending order. The bottom pane displays the results of the query as a table with 10 rows.

```
SELECT l.loan_id, l.loan_type, l.loan_amount, l.loan_status, b.branch_name
FROM loans l
JOIN branches b ON l.branch_id = b.branch_id
WHERE b.city = 'Lagos'
ORDER BY l.loan_amount DESC;
```

loan_id	loan_type	loan_amount	loan_status	branch_name
LN00123	personal	10000000	Active	Victoria Island
LN00351	SME	10000000	Active	Lekki
LN00371	Personal	10000000	Defaulted	Ikeja
LN00457	SME	10000000	Fully Paid	Ikeja
LN00337	SME	10000000	Active	Ikeja
LN00488	sme	10000000	Fully Paid	Victoria Island
LN00003	Business	10000000	Fully Paid	Lekki
LN00184	SME	10000000	Defaulted	Victoria Island
LN00397	personal	10000000	Active	Victoria Island
LN00434	personal	10000000	Defaulted	Lekki

Q4: Total Loans per Branch with Default Rate

For each branch, return branch_name, city, total loans, total loan amount, number of defaulted loans, and default rate (rounded to 1 decimal place). Include only branches with more than 5 loans. Sort by default rate descending.



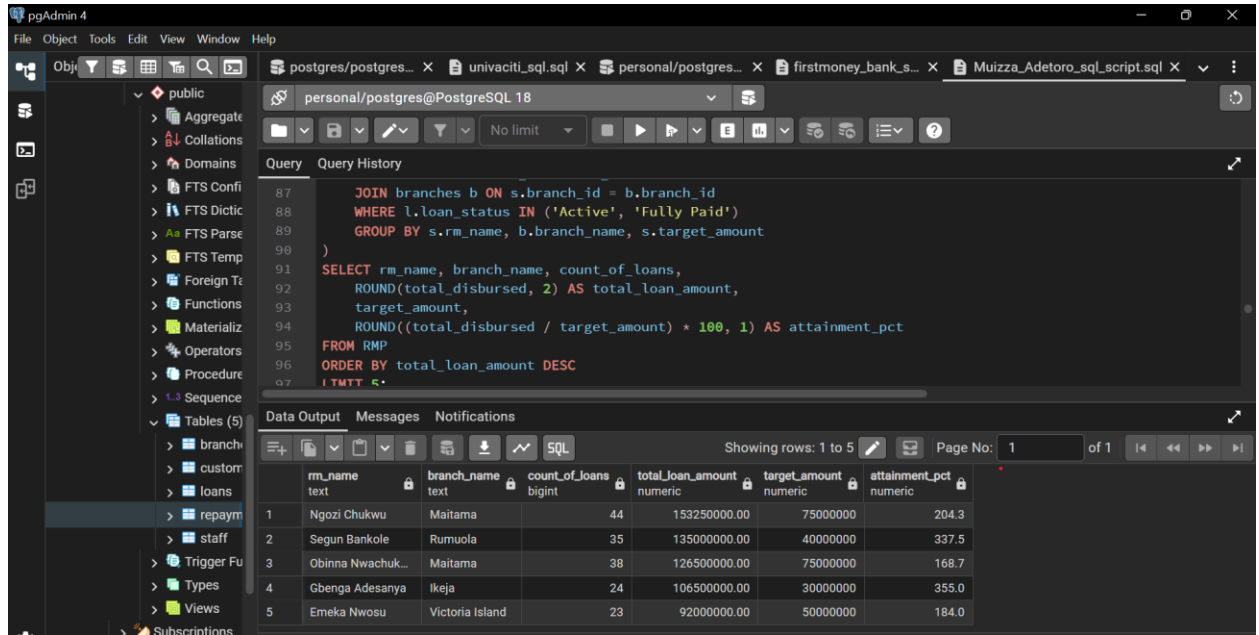
The screenshot shows the pgAdmin 4 interface. The left sidebar displays the database structure, including the 'loans' table. The central pane shows a SQL query that selects branch details and calculates the default rate. The bottom pane displays the query results in a table format.

```
SELECT branch_name, city, total_loans,
       ROUND(total_loan_amount, 2) AS total_loan_amount,
       no_defaulted_loans,
       ROUND((no_defaulted_loans * 100.0) / total_loans, 1) AS default_rate_pct
FROM LPB
WHERE total_loans > 5
ORDER BY default_rate_pct DESC;
```

	branch_name	city	total_loans	total_loan_amount	no_defaulted_loans	default_rate_pct
1	Sabon Gari	Kano	43	164500000.00	10	23.3
2	Wuse II	Abuja	26	81750000.00	5	19.2
3	Bompai	Kano	47	165500000.00	8	17.0
4	Port Harcourt G...	Port Harco...	36	93750000.00	6	16.7
5	Garki	Abuja	52	154750000.00	8	15.4
6	Victoria Island	Lagos	35	148500000.00	5	14.3
7	Maitama	Abuja	86	308750000.00	11	12.8
8	Lekki	Lagos	44	156750000.00	5	11.4

Q5: Top 5 Relationship Managers

Return the top 5 RMs by total loan amount (Active and Fully Paid only). Show rm_name, branch_name, count of loans, total loan amount, target_amount, and attainment_pct = (total disbursed ÷ target_amount) × 100, rounded to 1 decimal place. Sort descending.



The screenshot shows the pgAdmin 4 interface. The left sidebar displays the database structure, including tables like 'branches', 'loans', and 'repayments'. The central pane shows a SQL query that joins the 'loans' and 'branches' tables, filters for 'Active' and 'Fully Paid' loan statuses, and calculates the total loan amount and attainment percentage for each relationship manager (RM). The query is sorted by total loan amount in descending order.

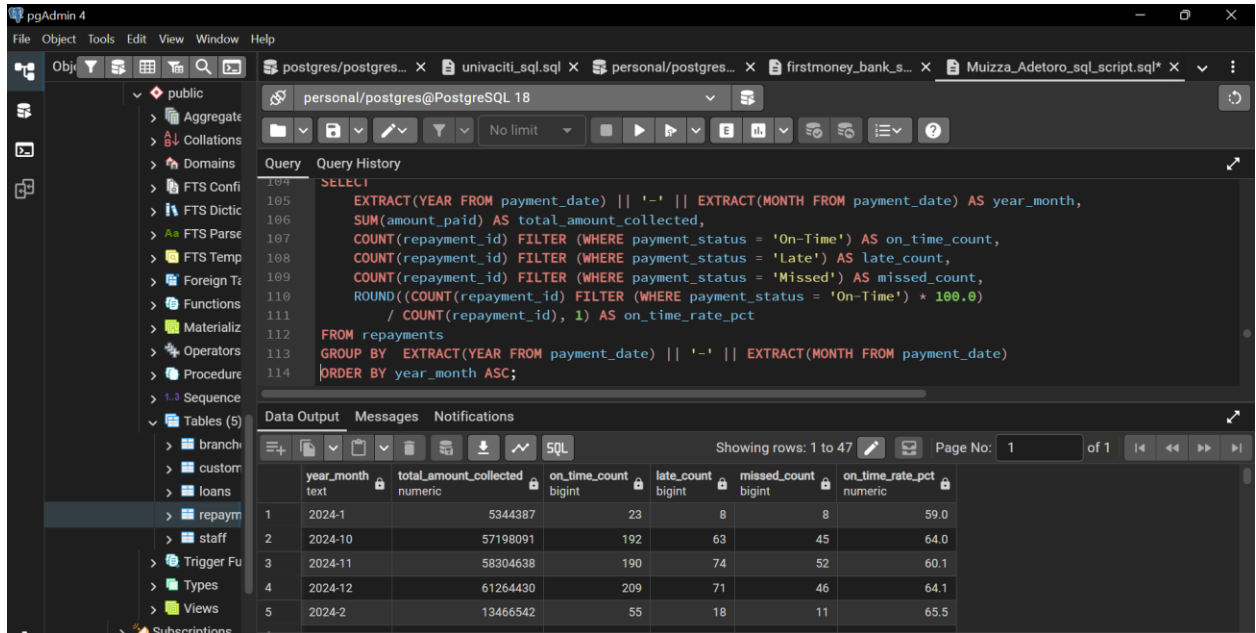
```
87 JOIN branches b ON s.branch_id = b.branch_id
88 WHERE l.loan_status IN ('Active', 'Fully Paid')
89 GROUP BY s.rm_name, b.branch_name, s.target_amount
90 )
91 SELECT rm_name, branch_name, count_of_loans,
92        ROUND(total_disbursed, 2) AS total_loan_amount,
93        target_amount,
94        ROUND((total_disbursed / target_amount) * 100, 1) AS attainment_pct
95 FROM RMP
96 ORDER BY total_loan_amount DESC
97
```

The bottom pane displays the results of the query, showing the top 5 RMs by total loan amount. The results are as follows:

	rm_name	branch_name	count_of_loans	total_loan_amount	target_amount	attainment_pct
1	Ngozi Chukwu	Maitama	44	153250000.00	75000000	204.3
2	Segun Bankole	Rumuola	35	135000000.00	40000000	337.5
3	Obinna Nwachuk...	Maitama	38	126500000.00	75000000	168.7
4	Gbenga Adesanya	Ikeja	24	106500000.00	30000000	355.0
5	Emeka Nwosu	Victoria Island	23	92000000.00	50000000	184.0

Q6: Monthly Repayment Trend

Using the repayments table, produce a monthly summary showing year-month (YYYY-MM), total amount collected, count of On-Time / Late / Missed payments, and on-time rate (rounded to 1 decimal place). Sort by year-month ascending.



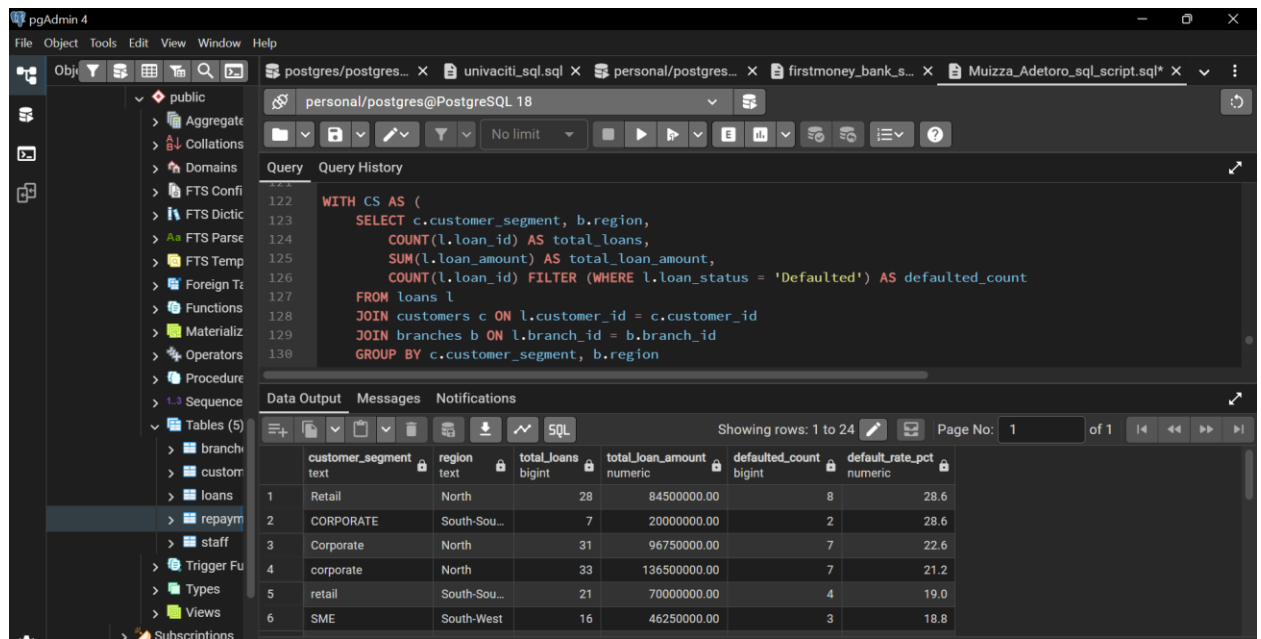
The screenshot shows the pgAdmin 4 interface. The left sidebar displays the database structure, including a table named 'repayments'. The central pane shows a SQL query that calculates a monthly summary of repayments. The query uses the 'repayments' table and filters for 'On-Time', 'Late', and 'Missed' payment statuses. The results are displayed in a table at the bottom, showing the year-month, total amount collected, and counts for on-time, late, and missed payments, along with the on-time rate percentage.

```
SELECT
    EXTRACT(YEAR FROM payment_date) || '-' || EXTRACT(MONTH FROM payment_date) AS year_month,
    SUM(amount_paid) AS total_amount_collected,
    COUNT(repayment_id) FILTER (WHERE payment_status = 'On-Time') AS on_time_count,
    COUNT(repayment_id) FILTER (WHERE payment_status = 'Late') AS late_count,
    COUNT(repayment_id) FILTER (WHERE payment_status = 'Missed') AS missed_count,
    ROUND((COUNT(repayment_id) FILTER (WHERE payment_status = 'On-Time') * 100.0)
          / COUNT(repayment_id), 1) AS on_time_rate_pct
FROM repayments
GROUP BY EXTRACT(YEAR FROM payment_date) || '-' || EXTRACT(MONTH FROM payment_date)
ORDER BY year_month ASC;
```

	year_month	total_amount_collected	on_time_count	late_count	missed_count	on_time_rate_pct
1	2024-1	5344387	23	8	8	59.0
2	2024-10	57198091	192	63	45	64.0
3	2024-11	58304638	190	74	52	60.1
4	2024-12	61264430	209	71	46	64.1
5	2024-2	13466542	55	18	11	65.5

Q7: Customer Segment Default Analysis (3-table JOIN)

Join loans, customers, and branches. For each combination of customer_segment and region, return total loans, total loan amount, defaulted loan count, and default rate (%). Include only combinations with at least 3 loans. Sort by default rate descending.



The screenshot shows the pgAdmin 4 interface. The left sidebar displays the database structure, including the 'public' schema and various objects like Aggregate, Collations, Domains, FTS Config, FTS Dictic, FTS Parse, FTS Temp, Foreign Tables, Functions, Materialized Views, Operators, Procedures, Sequences, Tables (5), and Subscriptions. The 'Tables' section is expanded, showing 'branch', 'customer', 'loans', 'repaym', 'staff', 'Trigger Functions', 'Types', and 'Views'.

The main window displays a SQL query in the 'Query' tab. The query is as follows:

```
WITH CS AS (
  SELECT c.customer_segment, b.region,
         COUNT(l.loan_id) AS total_loans,
         SUM(l.loan_amount) AS total_loan_amount,
         COUNT(l.loan_id) FILTER (WHERE l.loan_status = 'Defaulted') AS defaulted_count
  FROM loans l
  JOIN customers c ON l.customer_id = c.customer_id
  JOIN branches b ON l.branch_id = b.branch_id
  GROUP BY c.customer_segment, b.region
)
```

The 'Data Output' tab shows the results of the query. The results are displayed in a table with the following columns: customer_segment, region, total_loans, total_loan_amount, defaulted_count, and default_rate_pct. The results are sorted by default_rate_pct in descending order.

customer_segment	region	total_loans	total_loan_amount	defaulted_count	default_rate_pct
Retail	North	28	84500000.00	8	28.6
CORPORATE	South-Sou...	7	20000000.00	2	28.6
Corporate	North	31	96750000.00	7	22.6
corporate	North	33	136500000.00	7	21.2
retail	South-Sou...	21	70000000.00	4	19.0
SME	South-West	16	46250000.00	3	18.8

-Add a one-sentence comment in your .sql file below Q7 explaining what the result reveals.

The result reveals that the Retail segment in the North and Corporate segment in the South-South have the highest default rates, suggesting that the bank needs to investigate regional economic factors affecting these specific groups.

Alter Table and Column

I altered the 'loan_amount', 'target_amount', 'amount_paid', 'payment_date' column from TEXT to NUMERIC. This was necessary to perform aggregations (SUM/AVG) and to ensure financial data integrity.

